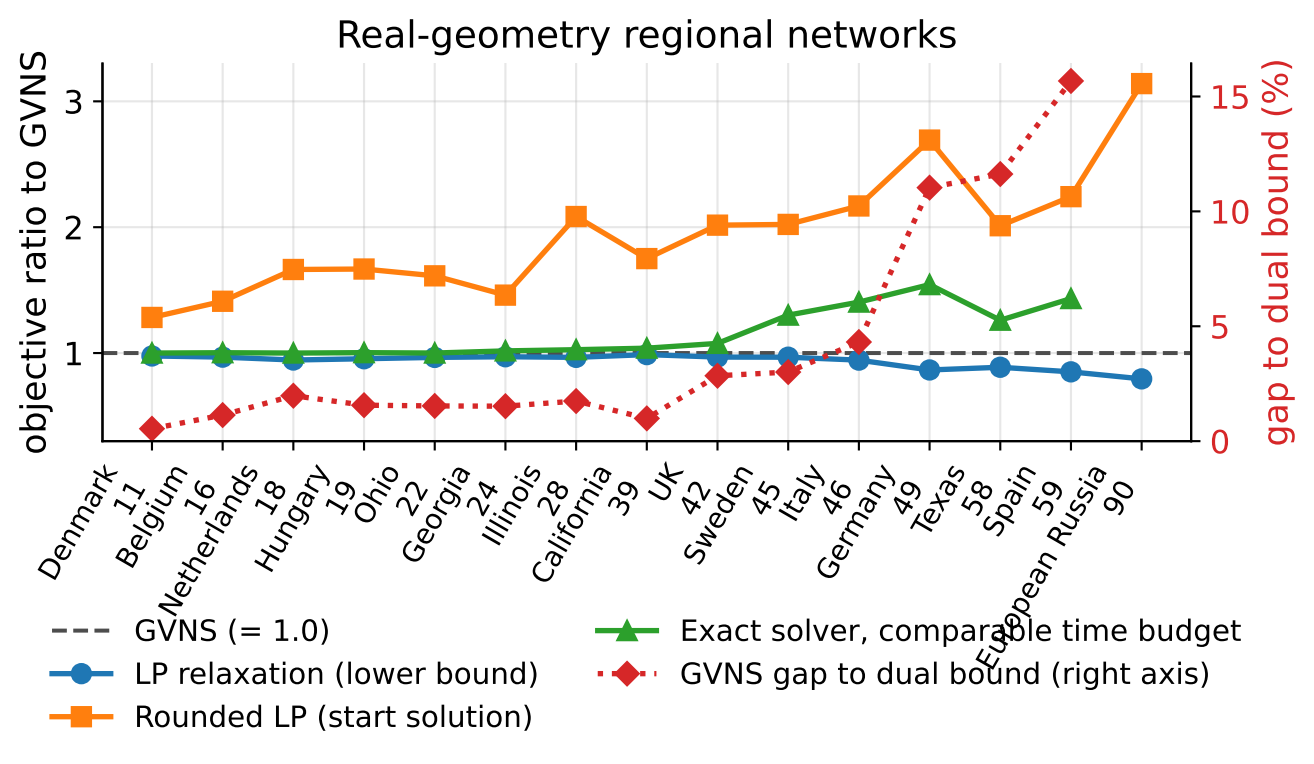
Real-geometry regional networks
objective ratio to GVNS
gap to dual bound (%)
3
2
1
15
10
5
0
Denmark 11
Belgium 16
Netherlands 18
Hungary 19
Ohio 22
Georgia 24
Illinois 28
California 39
UK 42
Sweden 45
Italy 46
Germany 49
Texas 58
Spain 59
European Russia 90
GVNS (= 1.0)
LP relaxation (lower bound)
Rounded LP (start solution)
Exact solver, comparable time budget
GVNS gap to dual bound (right axis)